

Chapter 22: Deception at a Cost: Reinforcement Learning for Resilient Infrastructure

Dr. Tapashi Kashyap Das

Assistant Professor

Discipline of Computer Science

Krishna Kanta Handiqui State Open University (KKHSOU)

Guwahati, Assam, India

tapashi.kashyap@gmail.com

Abstract

Critical infrastructure systems face attacks that increasingly rely on deception to overload defenses and trigger cascading failures. A recent quantitative framework treats deception as an explicit, cost-aware strategic resource and shows that moderate, calibrated decoy deployment can improve resilience per unit of defensive expenditure when the attacker follows a fixed, rule-based strategy. This assumption becomes harder to sustain as adversaries adopt automated reconnaissance and learning-based tactics. This chapter reviews that cost-aware framework and examines how reinforcement learning, adaptive decoy allocation, and digital-twin-based simulation could be incorporated into it to preserve the efficiency of deception against adversaries capable of adapting their behavior. The discussion closes with guidance for critical infrastructure operators on allocating resources between static and adaptive deception under budget constraints, together with a set of concrete steps for testing the proposed extensions empirically.

Keywords: *deceptive defense, critical infrastructure resilience, artificial intelligence, reinforcement learning, cost-aware security, moving target defense, digital twins*

1.1 Introduction

Critical infrastructure, including power grids, water treatment plants, transportation networks, and industrial control systems, has become progressively dependent on networked, software-defined control layers. This dependence has widened the attack surface available to adversaries well beyond the boundaries that traditional physical-security thinking was built to cover [1, 2]. Attackers rarely rely on force alone. Reconnaissance, false signalling, and manipulation of operator attention are used to fragment coordination among defenders and to precipitate cascading failure across interdependent subsystems [3–5]. Figure 1.1 situates these three claims visually: a representative physical asset, the digital control layer increasingly overlaid on it, and the adversarial activity this convergence exposes it to.



(a) Physical infrastructure asset (b) Digital-physical convergence (c) Adversarial engagement

Figure 1.1: Illustrative visual summary of the chapter's core setting: (a) a physical critical infrastructure asset, (b) its convergence with digital control and security layers, and (c) adversarial engagement with that infrastructure.

The three panels together summarize why deception has become a viable strategic resource for defenders rather than only for attackers. The physical scale and interdependence visible in Figure 1.1a explain why direct reinforcement of every asset is costly; the convergence shown in Figure 1.1b explains why that same infrastructure is newly reachable through digital channels; and the adversarial behavior in Figure 1.1c motivates modelling deception as a cost-aware defensive strategy in the sections that follow.

The defensive side of this contest is changing as well. Machine learning is now used for anomaly detection in supervisory control and data acquisition (SCADA) telemetry, for predictive maintenance, and, increasingly, for generating and managing deceptive artifacts such as honeypots, decoy data streams, and simulated system states that can be varied in response to observed attacker behavior [6,7]. At the same time, automated reconnaissance tools and adaptive malware are lowering the cost of mounting sophisticated intrusion attempts, so the balance of advantage between attacker and defender in a deception-based contest can no longer be assumed static.

Most quantitative treatments of deception in critical infrastructure defense restrict attention to non-learning, rule-based adversaries. A recent cost-aware resilience modelling framework introduced by the present author follows this convention deliberately, in order to isolate the strategic trade-offs among defender strength, coordination, and deception without the added complexity of an adapting opponent [8]. The restriction is reasonable for a first exploratory model, and the resulting findings, summarized in Section 3, are robust under wide parameter variation. The restriction nonetheless describes a single point in a much larger design space, the point at which the attacker cannot learn from experience.

This chapter has two purposes. The first is to summarize the cost-aware deceptive defense framework of [8], a discrete-time agent-based model in which authentic defenders and low-cost decoys interact with a rule-based attacker under an explicit cost-resilience metric, treating it as a validated baseline rather than a finished answer. The second, and more central, purpose is to ask how the cost-resilience trade-offs identified under that baseline would change once the attacker, the defender, or both are permitted to learn. The argument developed here is that artificial intelligence changes the economics of deception rather than simply adding a new defensive tool to an existing list, and that the cost-aware resilience metric proposed in the baseline study should therefore be reconceived as a quantity that a controller updates continuously rather than one that is fixed once at design time.

The remainder of the chapter proceeds by asking what level of deception investment is cost-optimal when the attacker is fixed and how robust that optimum is (Section 3), how the optimum would shift once the attacker can learn from repeated engagements (Section 4.1), whether decoy allocation itself can be automated through reinforcement learning or Bayesian optimization and what this requires of the surrounding information and communication technology (ICT) infrastructure (Sections 4.2 to 4.4), what a combined architecture for AI-augmented deception looks like and what computational and governance questions it raises (Section 5), and finally what practical guidance follows for operators working under budget constraints, together with what remains open for future study (Sections 6 and 7).

1.2 Background and Related Work

Resilience research in critical infrastructure has traditionally emphasized robustness, redundancy, and rapid recovery. Foundational studies of network topology examined how failures propagate through interdependent systems and identified the structural conditions under which a small initial disturbance

produces cascading collapse [9,10]. Complementary work formalized resilience through recovery-trajectory metrics that quantify how quickly and completely a system restores performance after disruption [11, 12]. Game-theoretic treatments extended this line of work by modeling attacker and defender as strategic actors and by framing resource allocation as an optimization problem subject to budget constraints [13–15]. Control-theoretic contributions, including resilient and self-triggered model predictive control schemes, have addressed the maintenance of stable operation under active attack, particularly under false data injection targeting sensor measurements in cyber-physical systems [16–18].

Deception occupies a somewhat different position in this literature. Early treatments considered it an incidental feature of security operations, useful mainly as a byproduct of intrusion detection. Over roughly three decades it has instead become a deliberately engineered defensive resource, with dedicated surveys tracing its evolution from simple decoy hosts toward coordinated, systemic deception strategies in active cyber defense [19, 20]. Pawlick and Zhu develop a formal game-theoretic account of cyber deception that treats signaling and information asymmetry as first-class strategic variables rather than as auxiliary tactics [7], and practitioner-oriented accounts describe how deception reshapes defensive economics by altering what an attacker perceives and chooses to target, rather than by increasing raw defensive capacity [21]. Within critical infrastructure specifically, deception has been implemented through honeypots, decoy sensors, and controlled false data injection [22, 23], and it has been evaluated experimentally: the Tularosa study, for instance, used a controlled experiment with professional red teams to measure how deceptive elements change attacker decision times and error rates, providing some of the first quantitative human-subject evidence that deception measurably degrades attacker performance [24]. Deception has also been situated within broader moving-target-defense frameworks, which continuously reconfigure network parameters, addresses, or service locations to increase attacker uncertainty, and recent surveys have organized this literature explicitly around game-theoretic formulations [25–27].

Despite this substantial body of work, deception is rarely treated as an independent, continuously tunable strategic variable within quantitative, system-level resilience models. Most treatments instead position it as a qualitative or ancillary component layered on top of a resilience analysis built around redundancy and recovery [28]. The cost-aware framework summarized in Section 3 addresses this gap directly by introducing a resilience metric that links infrastructure survival probability to the cumulative cost of detection, engagement, and losses, which allows the efficiency of deception to be compared against the efficiency of conventional defensive investment on a common scale [8].

A separate and rapidly expanding literature addresses artificial intelligence at the detection layer, aimed at identifying deceptive manipulation of sensor data rather than at managing deception as a defensive resource. Spatio-temporal graph neural architectures have been used to detect and localize false or dummy data injection in power systems while accounting for incomplete knowledge of network topology on the attacker’s part [29], and state-space decomposition combined with sliding-mode observers has been used for real-time estimation of injected false data in load-frequency control loops [30]. These approaches optimize classification accuracy and detection latency at the signal layer and do not model the allocation of resources between direct defense and deception, nor do they consider an attacker capable of learning around a fixed deceptive posture.

A third strand of work treats artificial intelligence as an active participant within the deception process itself. Generative adversarial architectures have been proposed to synthesize convincing decoy traffic and

defeat reconnaissance attempts in industrial control systems [23], and reinforcement learning has been applied directly to network defense problems, including intrusion response and adaptive honeypot management, with deep reinforcement learning shown capable of learning defensive policies that outperform static rule sets in simulated network environments [31–33]. It is at the intersection of these three strands, cost-aware deception modeling, detection-layer artificial intelligence, and artificial intelligence as an active defender or attacker, that the present chapter is positioned. Table 1 summarizes this positioning.

Table 1.1: Positioning of the present chapter relative to related literature strands.

Table 1.1: Positioning of the present chapter relative to related literature strands.

Strand	Typical focus	Gap addressed here
Topology and cascading failure	Structural vulnerability, recovery metrics	Deception not modeled as a tunable resource
Game-theoretic moving target defense	Reconfiguration, equilibrium strategies	Cost-resilience trade-off not made explicit
Detection-layer artificial intelligence	Classification accuracy, detection latency	Attacker-defender resource allocation not modeled
Cost-aware deception baseline	Explicit cost-resilience metric, decoy economics	Attacker assumed non-learning and rule-based
Present chapter	AI-augmented, adaptive deception	Connects the static cost-aware baseline to learning agents

1.3 Recap: The Cost-Aware Deceptive Defense Model

This section summarizes the discrete-time, agent-based simulation framework introduced in [8], which functions as the empirical baseline for the artificial-intelligence extensions proposed in Sections 4 and 5.

1.3.1 Model Structure

The framework represents a stylized critical infrastructure protection scenario in which a central asset, for example a SCADA server or a substation controller, is protected by a ring of authentic defenders and, in some configurations, a set of low-cost decoys, against a single rule-based attacker advancing toward the infrastructure core. Each defender carries a fixed base strength and gains additional effectiveness through a coordination multiplier when several defenders engage the attacker together. Decoys carry much lower base strength but a wider detection radius, which makes them disproportionately visible and attractive to the attacker relative to their cost. The attacker follows a fixed priority rule: any detected decoy within a threshold range is engaged preferentially, and the attacker otherwise advances directly on the infrastructure. The attacker’s strength is depleted through successive engagements with defenders and decoys alike.

Resource expenditure accumulates as a cost metric, incremented whenever a defender or decoy detects or engages the attacker. This produces a cost-aware resilience indicator, survival probability normalized by

cumulative defensive cost, that is designed specifically to capture deception as an economic resource. Classical resilience indicators built around recovery trajectories, and cybersecurity indicators such as mean time to detect or mean time to compromise, do not capture this economic dimension, since they measure timing or restoration rather than the expenditure required to achieve a given level of survivability [12].

Figure 1.2 summarizes the discrete-time workflow underlying each simulation run. Each run begins with initialization of defenders, decoys, and the attacker, after which the system iterates through three phases. In the detection phase, defenders and decoys assess the attacker's proximity, and any successful decoy detection immediately reduces attacker strength while providing an intelligence bonus to authentic defenders. In the engagement phase, defenders attempt to neutralize the attacker while decoys impose additional penalties if engaged. In the update phase, the attacker advances toward its selected target and its strength evolves according to the rules described above. After each cycle, the simulation checks whether the infrastructure has been breached, whether the attacker has been defeated, whether all defenders have been disabled, or whether the maximum time horizon has been reached, and terminates when any of these conditions holds, recording survival probability, mean defense cost, and cost-effectiveness as outputs.

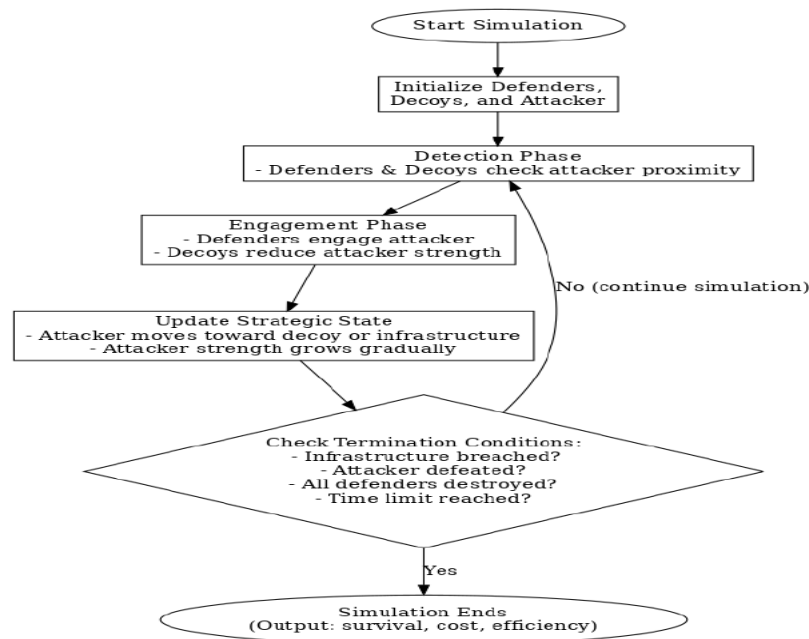


Figure 1.2: Discrete-time workflow of the cost-aware deceptive defense simulation. The system iterates through detection, engagement, and strategic update phases until one of four termination conditions is met, after which survival probability, defense cost, and efficiency metrics are recorded. Adapted from the baseline framework in [8].

1.3.2 Key Findings

Four findings from the baseline model carry particular weight for the artificial-intelligence extension proposed later in this chapter.

Survival probability increases monotonically with defender count, yet cost-effectiveness follows a non-monotonic pattern that peaks at an intermediate density, twelve defenders in the reported baseline configuration, beyond which additional defenders raise cumulative cost without a proportional gain in resilience. Deception behaves similarly rather than linearly: two decoys already provide reliable and fairly uniform survival gains across defender densities, four to six decoys produce the largest improvements at intermediate defender densities, and beyond this range additional decoys yield diminishing returns as survival approaches saturation near unity. The optimal intensity of deception is also scale dependent. As the number of defenders grows from four to thirty-two across the scalability analysis reported in the baseline study, the decoy-to-defender ratio that maximizes cost effectiveness falls from roughly one quarter at four defenders to roughly three hundredths at thirty-two defenders, indicating that deception contributes most where direct defensive redundancy is scarce and contributes comparatively little once redundancy is already high. Sensitivity analysis across wide parameter ranges, varied independently by plus or minus forty percent, confirms that these qualitative trends persist and that coordination and engagement radius dominate cost-effectiveness relative to deception-related parameters. This robustness, however, has only been established under a fixed, non-adaptive attacker, and the original study identifies adaptive or learning adversaries explicitly as its principal direction for future extension.

Figure 1.3 illustrates three of these patterns directly: the effect of decoy count on survival across defender densities, the corresponding cost-effectiveness of decoy count at representative defender densities, and the effect of network scale on survival under varying decoy-to-defender ratios. The defender-density cost-effectiveness optimum described above, peaking at twelve defenders in the baseline configuration, is reported in the original simulation study and is not reproduced as a separate panel here.

This last point sets up the remainder of the chapter. Robustness under parameter perturbation and robustness under an adversary capable of observing outcomes and revising its own strategy are different properties, and only the first has so far been demonstrated. Sections 1.4 and 1.5 turn this distinction into a concrete design and research agenda.

1.4 AI/ICT Integration Layer: Conceptual Extension

1.4.1 Adaptive Adversaries via Reinforcement Learning

The baseline attacker follows a fixed rule that does not update in response to repeated encounters with the same decoy configuration. An artificial-intelligence-augmented adversary could instead be modeled as a reinforcement learning agent, using either value-based methods such as Q-learning or policy-gradient methods such as proximal policy optimization, with a state space that includes recent engagement outcomes and a reward function that depends on infrastructure proximity achieved and defensive resources depleted, net of the attacker's own strength lost [32–34]. Reinforcement learning of this kind has already been demonstrated to learn network-defense and network-attack policies that improve over successive episodes in simulated environments, which supports the plausibility of an attacker capable of similar learning within the ring-topology setting of the baseline model [31].

Cost-aware deceptive defense: summary of simulation results

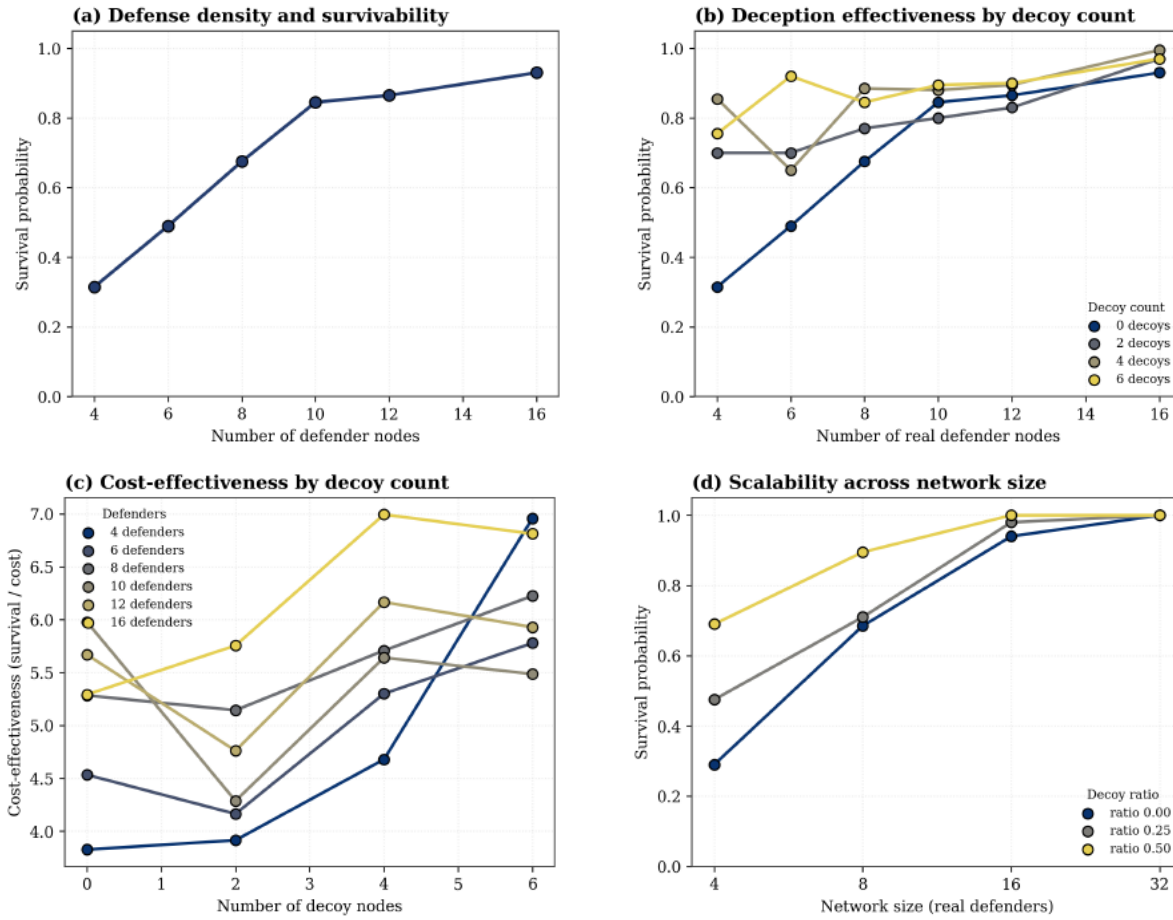


Figure 1.3: Summary of simulation results produced using the cost-aware deceptive defense framework introduced in [8], independently re-run for this chapter: (a) survival probability as a function of defender density, (b) the effect of decoy count on survival across defender densities, (c) cost-effectiveness as a function of decoy count for representative defender densities, and (d) survival probability across network scale under varying decoy-to-defender ratios.

Introducing such an agent suggests several qualitative shifts in the baseline findings that are worth testing directly. An attacker that learns across episodes to distinguish decoys from authentic defenders, for instance by learning that decoys rarely inflict meaningful strength loss even when engaged, would be expected to reduce the marginal survival benefit of a fixed decoy configuration over time, which would push the cost-effectiveness curves reported in Section 1.3 downward at any given decoy count. The point at which additional decoys stop paying for themselves, found at four to six decoys under the baseline's non-learning attacker, would likely arrive earlier under a learning attacker, since the informational advantage carried by any static decoy pattern degrades once the adversary has been exposed to it repeatedly. Once the attacker is capable of learning, the value of deception depends not on any single decoy configuration but on the rate at which decoys can be varied relative to the rate at which the attacker learns, which reframes deception from a design-time parameter into an ongoing control problem rather than a one-time allocation decision.

1.4.2 AI-Driven Dynamic Decoy Allocation

The natural response to a learning attacker is a learning defender. Rather than fixing decoy count and placement at design time, decoy allocation can be treated as an online decision problem in which a controller, at each interval, re-tunes decoy density, position, and detection radius based on recently observed attacker behavior, subject to the same cumulative cost constraint used in the baseline model. Bayesian optimization offers one route to this kind of online tuning when the interaction budget with a real or simulated attacker is limited, since it is designed to find good parameter settings using comparatively few evaluations [35], while reinforcement learning offers a route better suited to settings where longer training against a simulated attacker is feasible before deployment.

This reframes the baseline's cost-aware resilience metric,

$$CE(t) = \frac{P_{\text{survival}}(t)}{C_{\text{total}}(t)},$$

from a quantity evaluated once for a fixed configuration into a time-varying quantity that an adaptive controller seeks to maximize online,

$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t CE(t) \mid \pi \right],$$

where π denotes a decoy-allocation policy, γ a discount factor reflecting the operator's time preference, and the expectation is taken over the trajectory induced by the attacker, which may itself be learning. This formulation extends the baseline's static optimization, which seeks the single best fixed defender-decoy configuration, into a sequential decision problem that standard reinforcement learning methods are designed to solve, while keeping the original cost-aware resilience metric as the reward signal, so that the artificial-intelligence extension remains directly comparable to the validated baseline rather than replacing its evaluation criterion with a different one.

A controller of this kind could be trained within the same ring-topology abstraction used by the baseline model, treating the simulation as a low-cost training environment before any real-world deployment is attempted, which connects naturally to the digital twin concept discussed next.

1.4.3 Digital Twins for Predictive Resilience

The baseline simulation, being a discrete-time, parameterized, agent-based model with an explicit cost-aware output, is itself well suited to serve as the computational core of a digital twin of an actual critical infrastructure defensive posture. The digital-twin concept, originally developed for manufacturing systems as a virtual replica continuously synchronized with a physical asset [36], has since been extended to industrial systems generally [37] and, specifically, to cyber-physical security, where digital twins have been proposed as a means of testing and predicting the effect of security interventions before they are applied to a live system [38]. In this role, the model would be recalibrated continuously against telemetry from the environment it represents, such as real intrusion-detection alert rates or honeypot engagement logs, allowing operators to estimate the marginal cost-effectiveness of, for example, adding two additional decoys before committing real budget, using the reinforcement-learning-based policy search described above as the twin's internal optimizer.

This use case also implies a boundary that should be stated plainly. The digital twin role suits predictive, decision-support use, in which a human operator reviews the twin's recommendations before acting on them, rather than fully autonomous, closed-loop control of live defensive posture without oversight, given the safety-critical nature of the infrastructure the twin represents [38].

1.4.4 ICT Infrastructure Considerations

Deploying artificial-intelligence-driven, adaptive deception in an operational technology environment introduces requirements that the stylized baseline model does not need to address. Decisions to re-allocate decoys within programmable logic controller networks or SCADA gateways may need to be computed at the network edge, close to the physical process, given the latency sensitivity of control loops in such environments [3]. Reinforcement learning methods typically require substantial interaction data to converge, and in a real critical infrastructure setting this data is scarce by definition, since large-scale successful attacks are rare events, which motivates pretraining any policy extensively within the simulation environment before attempting any limited fine-tuning against live data. Finally, an artificial-intelligence system that manages decoys dynamically becomes itself a target of some value, and its compromise could invert the intended asymmetry by revealing rather than concealing the location of authentic defensive assets, a risk that should be weighed explicitly against the efficiency gains the system is expected to deliver.

1.5 Proposed Hybrid Framework

Building on Sections 1.3 and 1.4, a three-layer architecture is proposed here that preserves the interpretability of the original cost-aware model while incorporating artificial intelligence-driven adaptivity. The simulation, or digital-twin, layer consists of the validated agent-based model of [8], recalibrated periodically against operational telemetry and used as a sandbox within which candidate policies are evaluated before deployment. The adaptive policy layer consists of a reinforcement-learning-based or Bayesian-optimization based controller that proposes decoy-allocation policies, trained primarily within the simulation layer and fine-tuned conservatively against live data once available. The operational deployment layer consists of the physical or cyber decoys and defenders themselves, instrumented to report engagement and cost telemetry back to the simulation layer, which closes the loop between prediction and observed outcome.

A central working hypothesis for this architecture, derived directly from Section 1.4.1, is that adaptive attackers reduce the marginal benefit of static decoys and that dynamic reallocation is required to sustain cost-effectiveness against them. This hypothesis is directly testable within the existing simulation framework, since it requires only that the fixed-rule attacker be replaced by a reinforcement learning agent and that the sensitivity and scalability analyses already reported in the baseline study be repeated under this new condition, which offers a concrete and comparatively low-cost first step toward empirical validation of the argument made in this chapter.

Computational and data-governance implications follow from the layered structure itself. The simulation layer requires no sensitive operational data and can be shared openly, in the same way the baseline model's implementation already is through its public repository. The adaptive policy layer requires access to aggregated and anonymized engagement telemetry rather than raw sensor data, which limits its exposure to privacy and security risk relative to a system with direct access to live operational data. The operational layer requires the strongest access controls of the three, since it mediates directly

between the artificial-intelligence system and live defensive infrastructure, and any weakness at this layer would undermine the protections built into the layers above it.

1.6 Implications for Practice

For operators allocating limited security budgets across defenders and decoys, the baseline model together with the extension proposed here supports a staged approach that depends on current defender density. Where defender density is low, coordination and engagement capacity should be strengthened before any substantial investment in deception, whether static or adaptive, since the baseline results show coordination and engagement radius dominating cost-effectiveness precisely in this regime. Where defender density is intermediate, deception investment yields its greatest expected return, and the baseline model suggests allocating roughly one quarter to one half of defender-equivalent resources to decoy mechanisms; this is also the regime in which adaptive allocation is likely to deliver the largest gain over a static configuration, since it is exactly where a learning attacker would otherwise erode the value of a fixed decoy pattern fastest. Where defender density is already high and survival probability approaches saturation, additional decoys of either kind yield diminishing returns, and resources are better directed toward maintaining existing redundancy or toward detection-layer methods such as the localization techniques used against false data injection [29, 30], rather than toward further expansion of deception.

From a governance perspective, standards and audit frameworks for critical infrastructure cybersecurity increasingly reference artificial-intelligence-augmented defense, and the cost-aware resilience metric developed in the baseline study offers an auditable quantity, survival probability per unit of defensive cost, that could be incorporated directly into compliance reporting, providing a link between the academic modeling presented here and regulatory practice. Operators should also weigh the possibility that an artificial intelligence- driven deception layer becomes an adversarial target in its own right, and should allocate resources to securing this layer in proportion to its operational criticality rather than to its nominal cost alone.

1.7 Limitations and Future Research Directions

The contribution of this chapter is conceptual and architectural rather than empirical. The artificial-intelligence extensions proposed in Section 1.4 have not yet been implemented or tested against the baseline model, and doing so is the most direct next step for future work. Several further limitations, inherited in part from the baseline study, deserve explicit mention. The underlying model uses a stylized, rotationally symmetric ring topology, whereas real critical infrastructure systems exhibit heterogeneous connectivity and irregular spatial layout, and extending the framework to grid, hub-and-spoke, or spatially clustered topologies, potentially informed by graph-based topology reconstruction methods developed for distribution systems under incomplete observability [39], would likely affect how artificial-intelligence-driven decoy allocation should be designed. The baseline study's sensitivity analysis varies parameters independently, and a full joint exploration across multiple parameters at once, particularly under an adaptive attacker, remains open.

Future work should also model an attacker capable of targeting the artificial-intelligence based decoy-allocation controller directly rather than only the decoys it manages, in order to assess whether the layered architecture proposed in Section 1.5 remains robust against this second-order threat. Finally, the hypotheses advanced in this chapter require testing on cyber-physical testbeds, or cautiously in live

environments with strong safety guarantees, before adaptive, artificial-intelligence-augmented deception could reasonably be recommended for operational deployment.

1.8 Conclusion

Deception, even in its simplest rule-based form, functions as a cost-effective complement to core defensive strength in critical infrastructure protection, with a well-defined and scale-dependent point of diminishing returns established under a non-adaptive attacker. This chapter has argued that artificial intelligence changes the underlying economics of that deception by threatening to erode the value of any fixed decoy configuration once adversaries gain the capacity to learn from repeated engagement, and that the natural response is to treat decoy allocation itself as an adaptive policy problem, layered on top of, and evaluated against, the existing cost-aware resilience framework rather than replacing it outright. The path from rule-based deception to adaptive, artificial-intelligence-augmented defense is accordingly best understood as a direct extension of the baseline framework rather than a departure from it: the same cost-aware resilience metric that anchors the original model can, with the additions described here, anchor its successor as well, offering both researchers and critical infrastructure operators a coherent basis for defending against adversaries that are themselves becoming more capable of learning.

Bibliography

- [1] A.A. Cardenas, S. Amin, S. Sastry, Secure control: towards survivable cyber-physical systems, in: Proc. 28th Int. Conf. Distributed Computing Systems Workshops, 2008, pp. 495-500.
- [2] J. Slay, M. Miller, Lessons learned from the Maroochy Water breach, in: Critical Infrastructure Protection, IFIP AICT vol. 253, Springer, 2008, pp. 73-82.
- [3] S. Sridhar, A. Hahn, M. Govindarasu, Cyber-physical system security for the electric power grid, Proc. IEEE 100(1) (2012) 210-224.
- [4] A. Teixeira, I. Shames, H. Sandberg, K.H. Johansson, A secure control framework for resource-limited adversaries, Automatica 51 (2015) 135-148.
- [5] J. Chen, J.S. Thorp, I. Dobson, Cascading dynamics and mitigation assessment in power system disturbances via a hidden failure model, Int. J. Electr. Power Energy Syst. 27(4) (2005) 318-326.
- [6] J. Inoue et al., Anomaly detection for a water treatment system using unsupervised machine learning, IEEE international conference on data mining workshops (ICDMW). IEEE, 2017.
- [7] J. Pawlick, Q. Zhu, Game Theory for Cyber Deception: From Theory to Applications, Springer, 2021.
- [8] K. Dutta, Cost-aware resilience modeling under deceptive attack scenarios in critical infrastructure defense, Int. J. Crit. Infrastruct. Prot. 53 (2026) 100843.
- [9] S.V. Buldyrev, R. Parshani, G. Paul, H.E. Stanley, S. Havlin, Catastrophic cascade of failures in interdependent networks, Nature 464 (2010) 1025-1028.
- [10] R. Albert, H. Jeong, A.L. Barabási, Error and attack tolerance of complex networks, Nature 406 (2000) 378-382.
- [11] E. Hollnagel, Resilience engineering and the built environment, Build. Res. Inf. 42(2) (2014) 221-228.
- [12] R. Francis, B. Bekera, A metric and frameworks for resilience analysis of engineered and infrastructure systems, Reliab. Eng. Syst. Saf. 121 (2014) 90-103.
- [13] G. Brown, et al., Defending critical infrastructure, Interfaces 36(6) (2006) 530-544.
- [14] L. Han, X. Zhao, Z. Chen, Y. Wu, X. Su, N. Zhang, Optimal allocation of defensive resources to defend urban power networks against different types of attackers, Int. J. Crit. Infrastruct. Prot. 35 (2021) 100467.

- [15] Q. Zhu, T. Başar, Game-theoretic methods for robustness, security, and resilience of cyberphysical control systems, *IEEE Control Syst. Mag.* 35(1) (2015) 46-65.
- [16] N. He, K. Ma, H. Li, H. Li, Resilient self-triggered model predictive control of discrete time nonlinear cyberphysical systems against false data injection attacks, *IEEE Intell. Transp. Syst. Mag.* 16(6) (2023) 23-36.
- [17] N. He, K. Ma, H. Li, Z. Fan, Adaptive input reconstruction based resilient MPC against deception attacks, *IEEE Trans. Ind. Electron.* 72(1) (2024) 938-948.
- [18] S. Liu, X. Wang, B. Niu, X. Song, H. Wang, X. Zhao, Adaptive resilient output feedback control against unknown deception attacks for nonlinear cyber-physical systems, *IEEE Trans. Circ. Syst. II* 71(8) (2024) 3855-3859.
- [19] L. Zhang, V.L.L. Thing, Three decades of deception techniques in active cyber defense, retrospect and outlook, *Comput. Secur.* 106 (2021) 102288.
- [20] W. Wang, Z. Lu, Cyber deception: overview and the road ahead, *IEEE Secur. Priv.* 16(2) (2018) 80-85.
- [21] A. Bushby, How deception can change cyber security defences, *Comput. Fraud Secur.* 2019(3) (2019) 12-14.
- [22] J.L. Rrushi, An exploration of defensive deception in industrial communication networks, *Int. J. Crit. Infrastruct. Prot.* 4 (2011) 66-75.
- [23] X. Qin, F. Jiang, X. Qin, L. Ge, M. Lu, R. Doss, CGAN-based cyber deception framework against reconnaissance attacks in ICS, *Comput. Netw.* 251 (2024) 110655.
- [24] K. Ferguson-Walter, T. Shade, A. Rogers, et al., The Tularosa study: an experimental design and implementation to quantify the effectiveness of cyber deception, in: *Proc. 52nd Hawaii Int. Conf. Syst. Sci.*, 2019.
- [25] S. Sengupta, A. Chowdhary, A. Sabur, A. Alshamrani, D. Huang, S. Kambhampati, A survey of moving target defenses for network security, *IEEE Commun. Surv. Tutor.* 22(3) (2020) 1909-1941.
- [26] J. Tan, H. Jin, H. Zhang, Y. Zhang, D. Chang, X. Liu, H. Zhang, A survey: when moving target defense meets game theory, *Comput. Sci. Rev.* 48 (2023) 100544.
- [27] E. Al-Shaer, Q. Duan, J.H. Jafarian, Random host mutation for moving target defense, in: *Security and Privacy in Communication Networks*, Springer LNICST vol. 106, 2013, pp. 310-327.
- [28] J.L. Johnson, Strategic culture: refining the theoretical construct, in: *Strategic Culture and Weapons of Mass Destruction*, Palgrave Macmillan, 2010, pp. 3-22.
- [29] Z. Qu, et al., Localization of dummy data injection attacks in power systems considering incomplete topological information, a spatio-temporal graph wavelet convolutional neural network approach, *Appl. Energy* 360 (2024) 122736.
- [30] C. Hong, Z. Liang, Y. Yang, P. Li, L. Chen, L. Bi, Y. Zhang, Research on detection and defense methods for false data injection attacks in power systems based on state-space decomposition, *Discov. Appl. Sci.* 7 (2025) 760.
- [31] T.T. Nguyen, V.J. Reddi, Deep reinforcement learning for cyber security, *IEEE Trans. Neural Netw. Learn. Syst.* 34(8) (2023) 3779-3795.
- [32] V. Mnih, K. Kavukcuoglu, D. Silver, et al., Human-level control through deep reinforcement learning, *Nature* 518 (2015) 529-533.
- [33] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, *arXiv preprint arXiv:1707.06347*, 2017.
- [34] R.S. Sutton, A.G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.

- [35] B. Shahriari, K. Swersky, Z. Wang, R.P. Adams, N. de Freitas, Taking the human out of the loop: a review of Bayesian optimization, Proc. IEEE 104(1) (2016) 148-175.
- [36] M. Grieves, Digital twin: manufacturing excellence through virtual factory replication, White paper, 2014.
- [37] F. Tao, H. Zhang, A. Liu, A.Y.C. Nee, Digital twin in industry: state-of-the-art, IEEE Trans. Ind. Inform. 15(4) (2019) 2405-2415.
- [38] M. Eckhart, A. Ekelhart, Digital twins for cyber-physical systems security: state of the art and outlook, in: Security and Quality in Cyber-Physical Systems Engineering, Springer, 2019, pp. 383-412.
- [39] S. Sun, et al., Integrated data-driven topology reconstruction and risk-aware reconfiguration for resilient power distribution systems under incomplete observability, Sci. Rep. 15 (2025) 30661.

Contributors/Authors:

- ^{1*} Dr. Tapashi Kashyap Das, Discipline of Computer Science, K K Handiqui State Open University, Guwahati, 781022, Assam, India
- ² Dr. Kishore Dutta, Department of Physics, Handique Girls' College, Guwahati, 781001, Assam, India

^{1*} Corresponding author email: tapashi.kashyap@gmail.com